



PEOPLE'S DEMOCRATIC REPUBLIC OF ALGERIA
Ministry of Higher Education and Scientific Research
University of Algiers1, Benyoucef Benkhedda
Faculty of science - CS Department



OPEN SOURCE COURSE

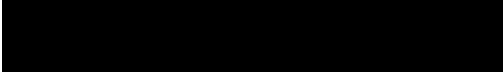


By : ABBADI MALIKA (M.A.B)
Academic year : 2025/2026

COURSE DESCRIPTION

- Learning Unit : Methodology
- Code : UEM11
- Open source course is an introductory course in CS program that initiate you to **IT concepts and tools** and **open source environment**.

CONTACT INFORMATION

- ▶ Title: OPEN SOURCE SOFTWARE
- ▶ University : Alger1, Benyoucef Benkhedda
- ▶ Faculty: science
- ▶ Department: Mathematics
- ▶ Instructor : ABBADI Malika
- ▶ For whom: L1 Section B students of Mathematics licence,
- ▶ Coefficient : 02
- ▶ Credit : 04
- ▶ Teaching hours volume : 21 h, 1.30h/week
- ▶ Evaluation : Exam
- ▶ Contact : 

COURSE OBJECTIVES

"Open Source course" aims to initiate you to IT and Internet

At the end of this course, you will be able to :

- *Identify IT and Open Source tools,*
- *Use technology effectively to analyze information and make decision,*
- *Compare software, OS, platforms based on their license,*
- *Contribute to open source projects and community (learn & teach)*



COURSE PLAN

Chapter 1 : InformationTechnology

- 1.1. Definitions

- 1.2. Tools

 - 1.2.1. Hardware

 - 1.2.2. Software

 - 1.2.3. Applications

Chapter 2 : Open source tools

Today topics : Chapter 1 – Part 1

InformationTechnology

1.1. Definitions

1.2. Tools : Hardware

1. Computers :

PC , Desktop computers, Laptops, servers, Workstations.

2. Peripherals :

Printers, Scanners, Keyboards, Mice, Camera, Sensors.

3. Mobile devices :

Smart-phones, tablets,

CHAPTER 1 :

Information Technology

What's information ?

What's data ?

What's technology ?

Weather forecast example

You receive some data about weather forecast in saharean city like Janet , Illizi.

The following messages come from a weather report.
Which conveys the most information, eliminate probabilities and reduce uncertainty?

1. The temperature will be 42 c°,
2. It will be sunny,
3. There will be a sandstorm,



Definitions

Example:

you receive some data about weather forecast in saharean city like Janet , Illizi.

The following messages come from a weather report. Which conveys the most information, eliminate probabilities and reduce uncertainty?

1. The temperature will be 42 c°, **good insight**
2. It will be sunny, **weak information**
3. There will be a sandstorm, **most informative**



Definitions

- **I**nformation, *n*. **D**ata communicated or received **processed** in some manner that resolves **uncertainty** about a particular fact or circumstance.
 - (In other words, information help you to make a decision)
 - Data, a set of facts resulting from measurment or observation of physical phenomenon.

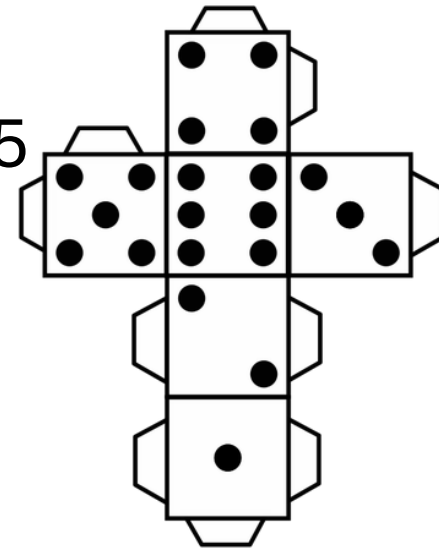
Another example

You receive some data about six-sided die roll.
Which conveys the most information, eliminate probabilities and reduce uncertainty?

Out of 6 how much possibilities can be eliminate



- A. The roll was not a 1
- B. The roll was a number less than 5
- C. The roll was not an even number
- D. The roll was a 3



Another example

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- (3) C. The roll was not an even number
- (5) D. The roll was a 3

Definitions

- **T**echnology, *n.* the application of scientific knowledge for practical purposes, especially in industry.
- Machinery and equipment developed from the application of scientific knowledge. (Oxford dictionary)

Information Technology

- Information Technology, refers to computing technologies like software, hardware, networking that enable **electronic data processing** and **communication**, and to **maintain electronic storage**.
- The main purpose of Information Technology is to help people perform their work **effectively** and achieve their objectives.

Let's add more: What does **CS** mean?

- **C**omputer **S**cience, is the science behind information technology. The study of computers and computing, including **theoretical** and **algorithmic foundations**, hardware and software creation and use for processing information. (Britanica)

Why don't say CS instead of IT

How many can you recognize



1-Computers

Computer : A **programmable** electronic device or machine that is capable of carrying out a sequence of arithmetic or logical operations in a defined manner , [Oxford Dictionary of computer science]

- containing **internal and external** components to receive, **process**, store and display **information**.

Computer type depends on internal capacity

There are several types of computers :

- Personal computers (PC), Desktop computers & Workstation,
- Laptops,
- Servers,
- Supercomputers,

PCs, Desktop computers& Workstations

- Personal computers are designed for general use (study, gaming, work from home, personal projects).
- The capacity is designed to be on-the-budget, cost effective, and large-market product line.
- This category include desktops, laptops, and tablets.



PCs, Desktop computers& Workstations

- **Desktops** are designed to be on desk (not portable). Mainly, for regular office use.
- Have separate external components mounted together like the screen, keyboard, mouse, printer, scanner.
- **Workstations** are for specific research/ scientific purpose, so they have higher capacity of calculation.



Laptops

- Laptops are a portable and smaller compact version of desktops, with all the components integrated into a single unit.
- Providing mobility, they are more suitable for travels and on-the-go work.
- They integrate a maximum of accessories ex: built-in webcams, Bluetooth and Wi-Fi.



Servers

- Servers are designed to manage and distribute resources and data to multiple users or devices. They are often used in businesses or organizations to store and share data and run applications
- The wider modern distributed version is the cloud (100, 1000, of servers).



Super computers

- Example :
Summit IBM supercomputer
- 9216 CPUs,
- 27648 GPUs NVIDIA,
- 2700 TeraB RAM = 2.63 PetaB.
- storage with 250 PB.



Super computers

- Supercomputers are extremely powerful computers composed of thousand of processors. that are designed to process data at extremely high speeds.
- They are often used for scientific research and other specialized applications (Big data, AI, cyber-security, cryptanalysis).



Mobile devices : smartphone, tablet

- **Smartphones** : A smart mobile phone is a programmable electronic hand-sized device, designed to ensure personal communications, connect to networks, and share data. Integrate a processor to run mobile operating systems (IOS, Android) and run small-sized programs (Apps) . Modern ones come with touchscreens .
- **Tablets** : are smaller than laptops but are larger than smartphones adapted for making calls, web surfing, creating and managing files.

Input/Output External Peripherals



- Keyboard
- Mouse
- Scanner



INPUT peripherals are hardware devices that send data to computer for processing

- Screen/monitor/display
- Printer



OUTPUT peripherals are hardware devices that receive/display/present data from computer

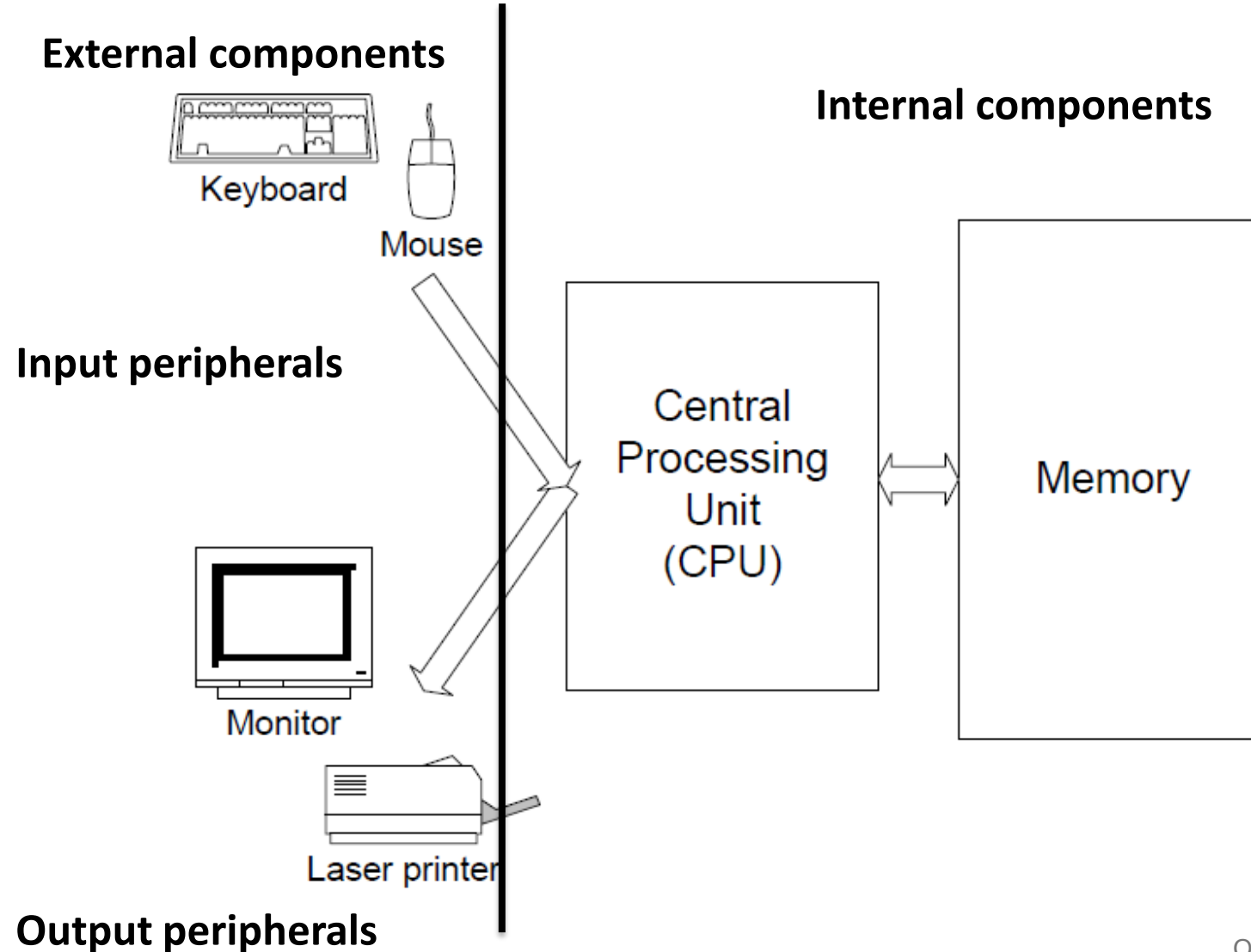
Today's topics

3. Computer internal components

4. Network devices : Router, switch, modems, optical fiber, Wi-Fi.

5. Smart ships : RFID, NFC, Embedded systems, Sensors (WSN & IoT (Internet of Things)).

Computer components

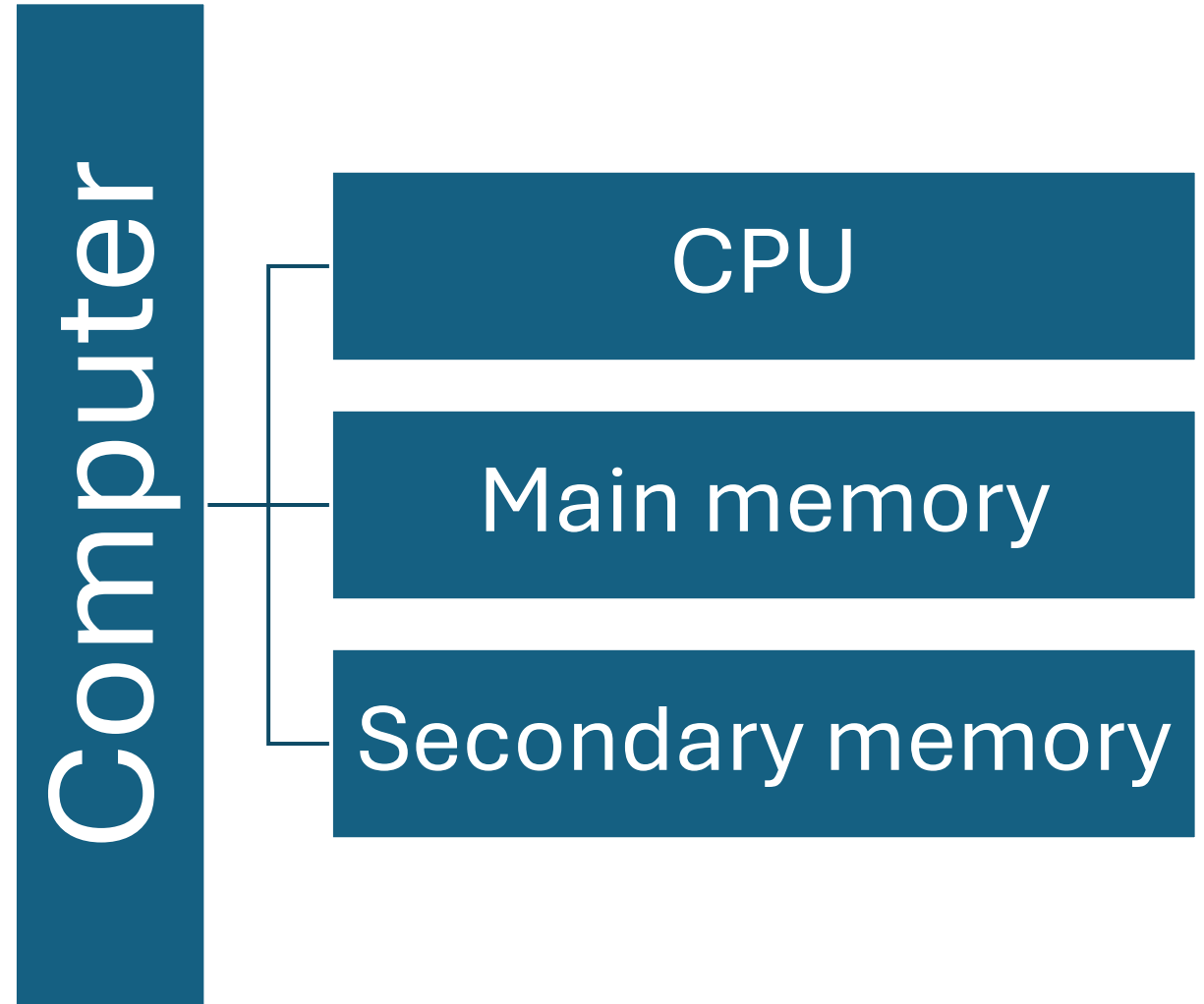


Internal Computer components

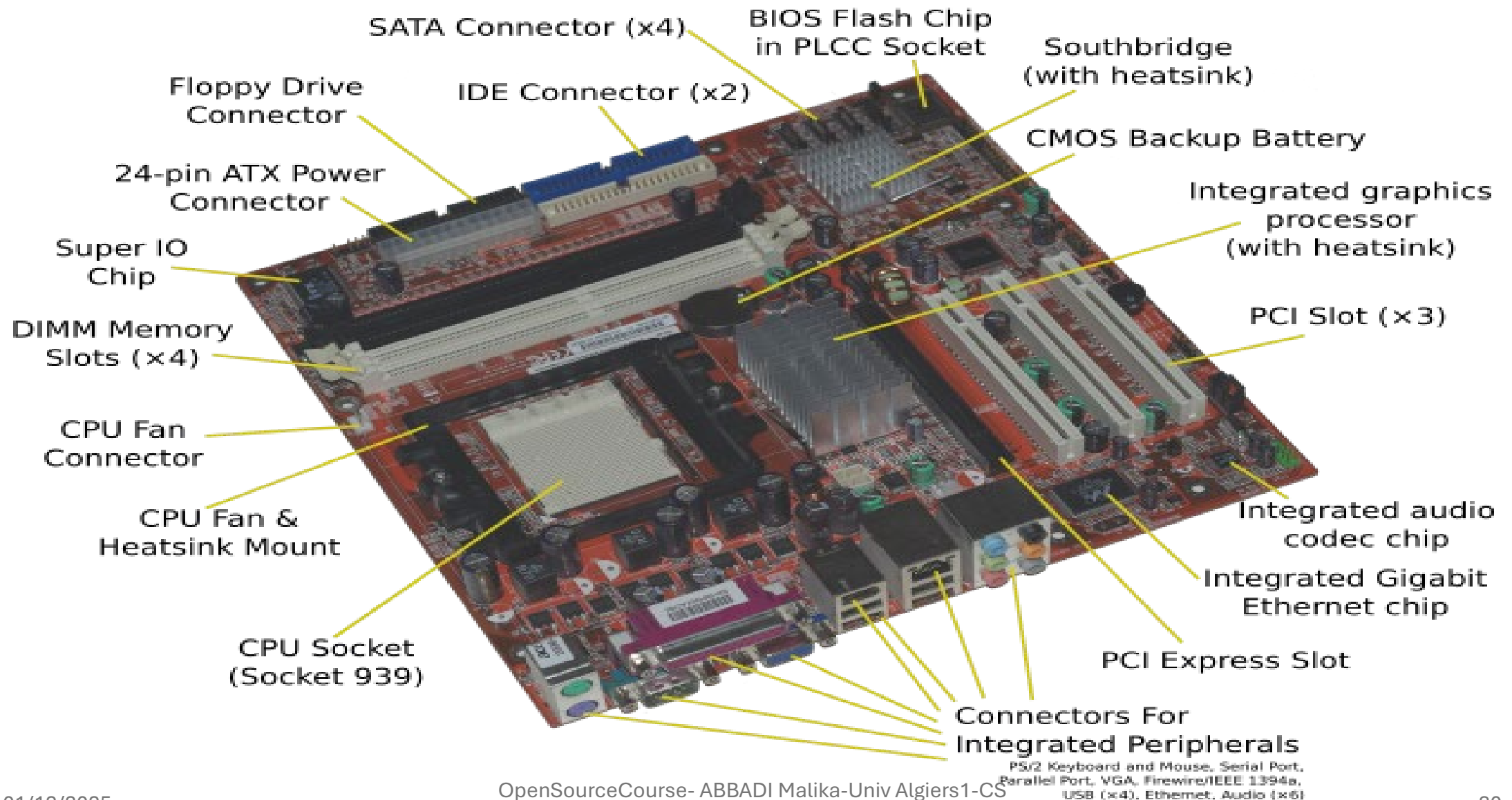
CPU : brain of a computer where calculations are made.

Main memory : for short time response data retrieval and transmission, with volatile data writing.

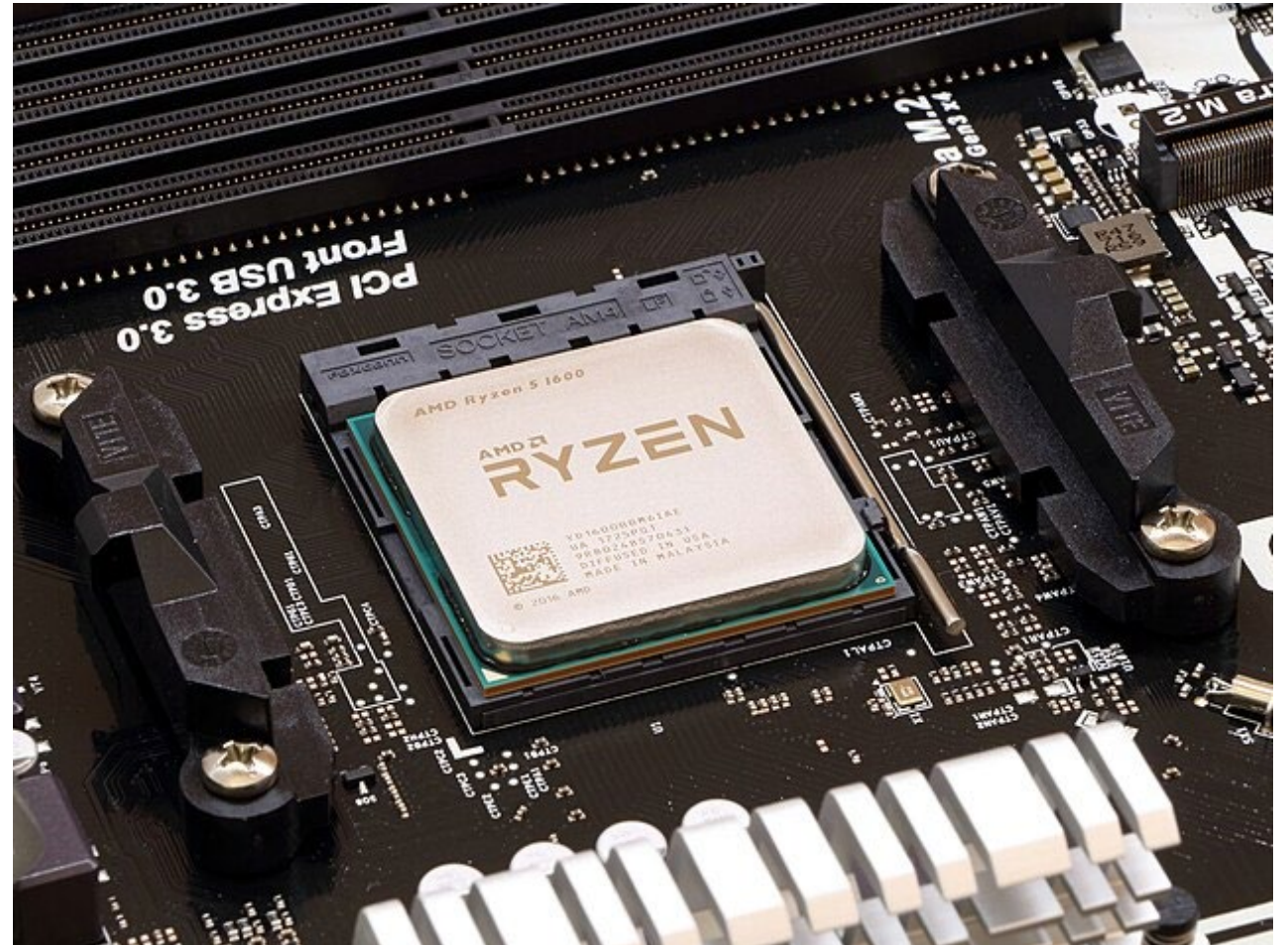
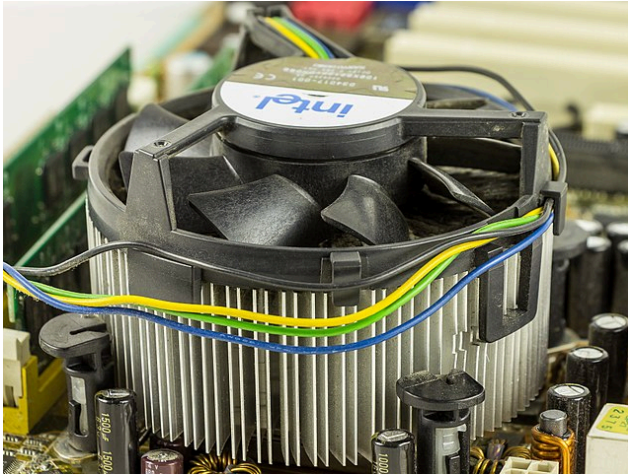
Secondary memory : is a storage device for non volatile long time storing of information and programs.



Motherboard



Processors – Central Processing Unit (CPU)



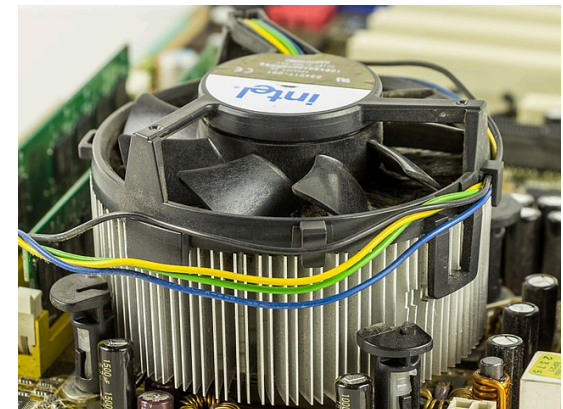
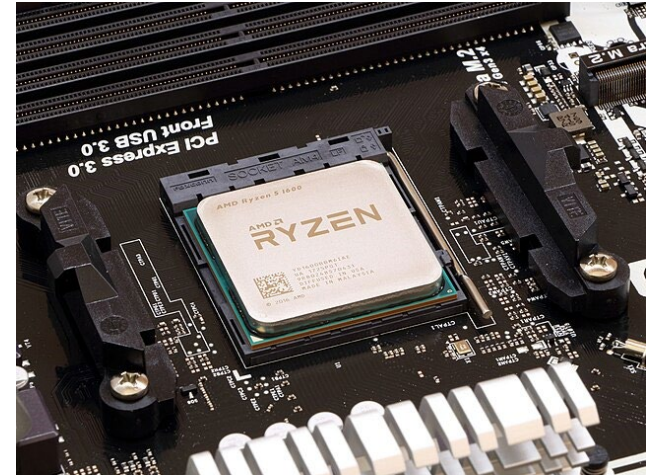
Processors – Central Processing Unit (CPU)

The CPU is usually placed in a special slot called a **socket** on the computer's **motherboard**.

Motherboard is the main circuit board that connects all the parts of a computer.

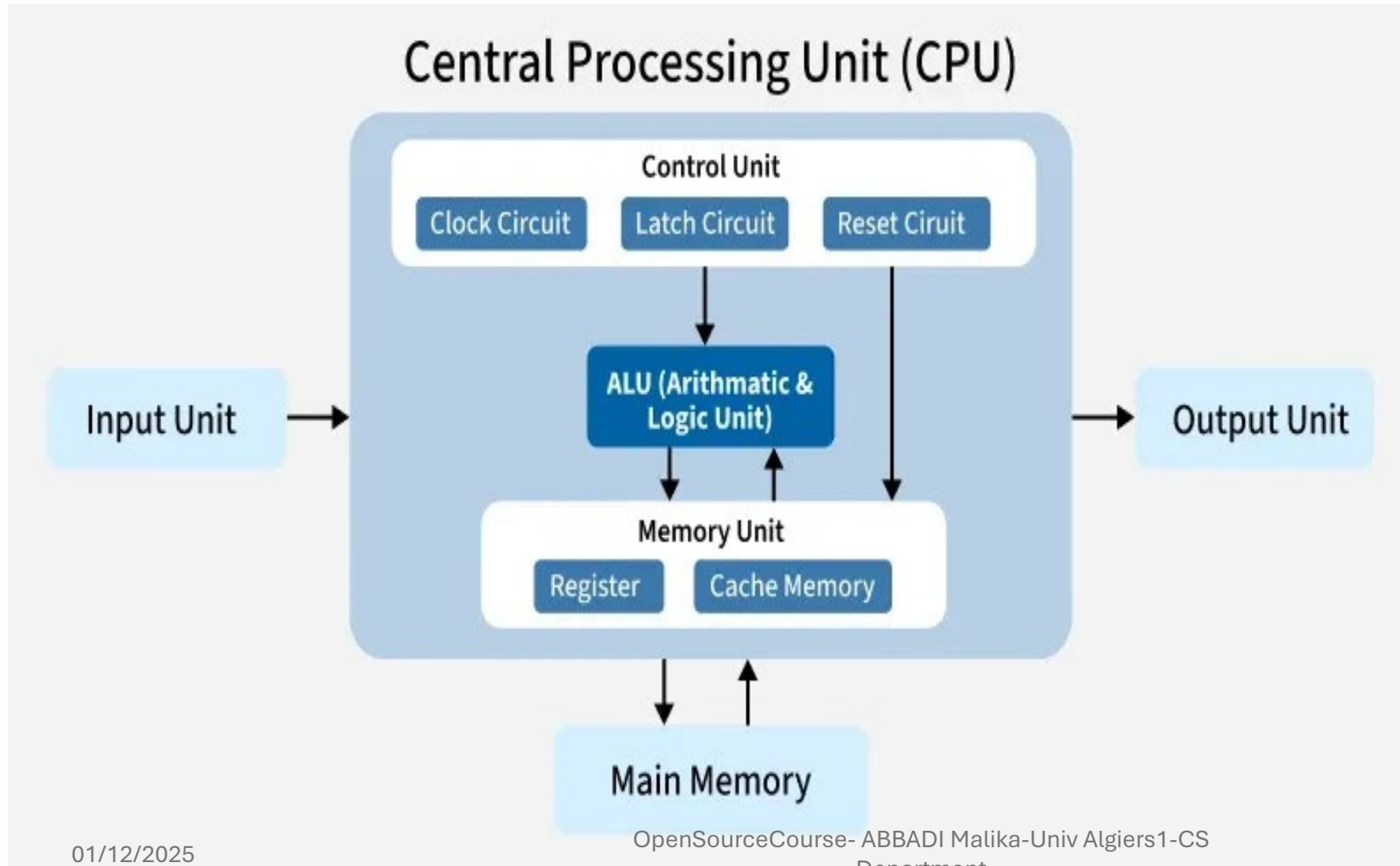
The CPU handles tasks like:

- Doing arithmetic / math calculations (like adding or multiplying numbers).
- Running apps or games.
- Helping input/output devices : keyboard, mouse, and screen to work/ communicate together.
- Storing and retrieving information when running tasks.



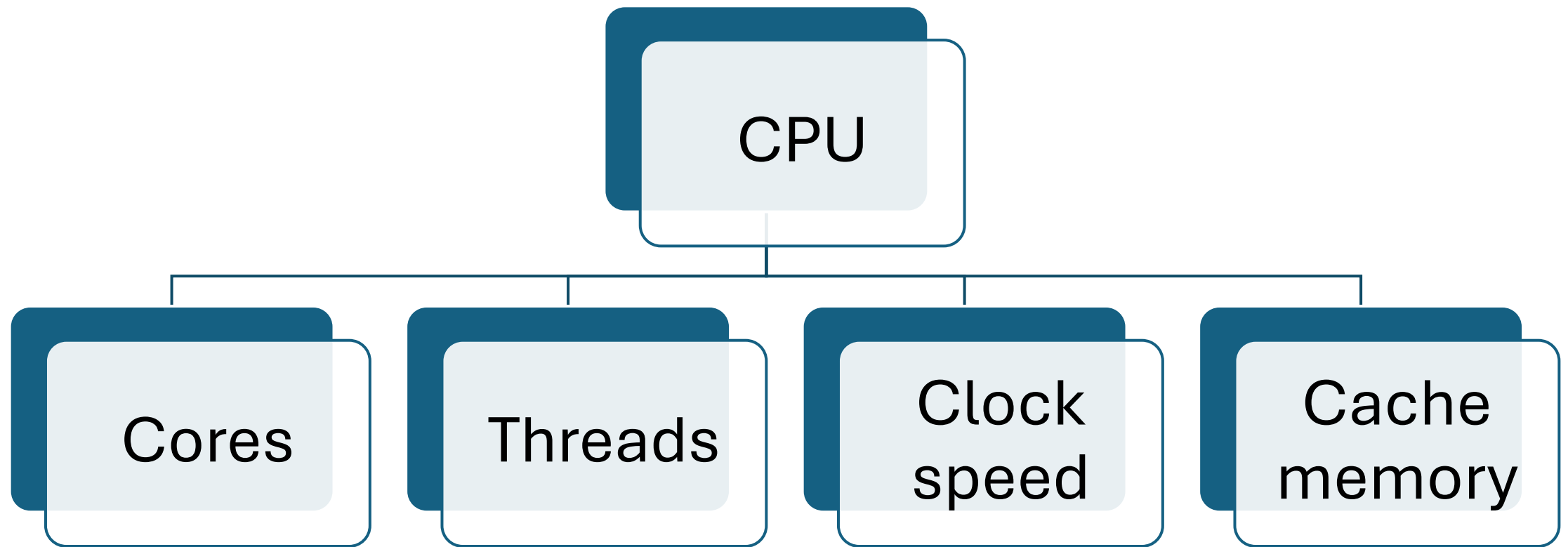
Cooling fan

CPU Anatomy- Von neumann architecture



- Control unit – Clock
- ALU unit (Arithmetic and logic)
- Memory unit (register, cache)

CPU – four Characteristics



CPU – 1st Characteristic- Core

- **Core:** for **multitasking**, Cores act as processors within the processor. Cores are processing units that read and carry out various program instructions. Processors are classified according to how many cores are included.
- CPUs with **multiple** cores can do **more tasks at once** and process instructions **faster** than single-core processors.

Types of CPUs

1. Single-Core CPU: The oldest type, used in the 1970s. It can only handle one task at a time, so it's slow for modern apps like games or web browsers.

- **1971:** Intel released the **Intel 4004**, the first-ever microprocessor (a **CPU on a single chip**) for PCs.



1. Dual-Core CPU: Has two cores, so it can handle two tasks at once. It's faster and better for multitasking, like listening to music while doing homework.

- **2005:** AMD introduced the first **dual-core processor**, allowing CPUs to **handle multiple tasks at once**.



Types of CPUs

- **Quad-Core CPU:** Has four cores, making it great for heavy tasks like **video editing** or playing modern **games**.
- Very fast and common in today's computers.
- The Intel Core 2 Extreme QX6700, named Kentsfield, released in November **2006**, is considered the first desktop quad-core.
- **2009:** Intel released the **Core i5**, a faster four-core processor.
- **And more : 8, 10 and over 20....(Intel)**



CPU – 2 & 3^d Characteristics

- **Threads:** for **multithreading** to allow running jobs concurrently and in parallel.
- Threads are the shortest sequences of programmable instructions that an operating system's scheduler can independently administer and send to the CPU for processing.
- Multithreading Hyper-threading is Intel's proprietary name.
- **Clock speed:** regulates the speed and frequency of computer operations. The clock manages the CPU's circuitry through the transmittal of electrical pulses. The delivery rate of those pulses is clock speed, which is measured in Hertz (Hz) or megahertz (MHz).

CPU – 4th characteristic : Cache memory

- Caches are storage areas near the register and arithmetic unit (ALU) to quickly access data in use.
- Caches store data in memory built into a CPU's processor chip to reach data retrieval speeds even faster than random access memory (RAM).
- There are three layers of cache L1, L2, L3
- Caches can be hardware or software components.

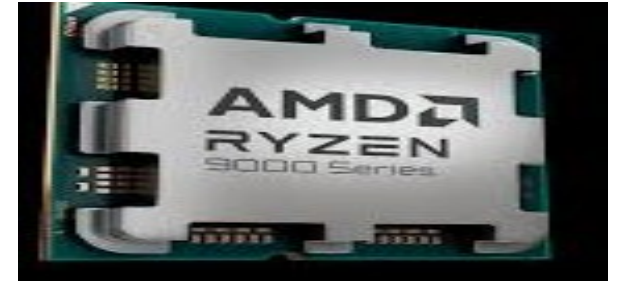


Main manufacturers



- **Intel (Integrated Electronics):** Intel launched processors and microprocessors through four product lines.
 - Intel Core, premium, are high-end line products.
 - Intel's Xeon® processors are targeted toward offices and businesses.
 - Intel's Celeron® and Intel Pentium® lines are considered slower and less powerful than the Core line.
- Intel processors use a CISC architecture (Complex Instruction Set Computer).
- There are two product's types: CPU, and NPU for Neural Processing Unit.

Main manufacturers : AMD

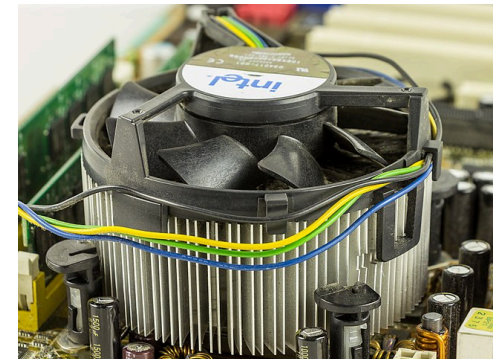


- **AMD (Advanced Micro Devices):** AMD processors follow a reduced instruction set computer (RISC) architecture.
- AMD's Ryzen™ processors are high-speed, high-performance microprocessors intended for the video game market. In the past, Athlon™ processors was considered AMD's high-end line, but now used as a basic computing alternative.
- AMD sells processors and microprocessors through two product types: CPUs and APUs (which stands for accelerated processing units). APUs are CPUs that have been equipped with proprietary Radeon™ graphics.

Intel Core i7 / i9 characteristics

Specification	Intel Core i7	Intel Core i9
Cores	4-8	6-18
Threads	8-16	12-36
Clock Speed	2.6-4.5 GHz	2.6-5.0 GHz
Cache	8-12 MB	12-19.25 MB

More capacity and capabilities mean high temperature more heating that need more cooling capacity



Graphics processing unit (GPUs)

- The GPU is a type of electronic circuit with immediate applications for PCs, smartphones and video game consoles, which was their original use. Its **increased speed is the cause of accelerating computer graphics.**
- Now, GPU is also adapted for purposes unrelated to graphics acceleration, like **cryptocurrency**, mining and the training of **neural networks.**

Random Access Memory (RAM)

RAM as **R**andom **A**ccess **M**emory, Stores data temporarily for quick access (write/read), while the computer is running

- Used by the Computer as the working area
- Holds the working program, the data being processed and the temporary results
- Volatile - contents are erased if power off
- Can be accessed randomly: can get any piece of data directly.
- Faster than permanent storage
- Not to be confused with ROM (Read-Only Memory)

ROM – Read Only Memory

- Retains data even without power
- Useful to store programs executed upon system start-up (e.g. BIOS)



Image from public domain

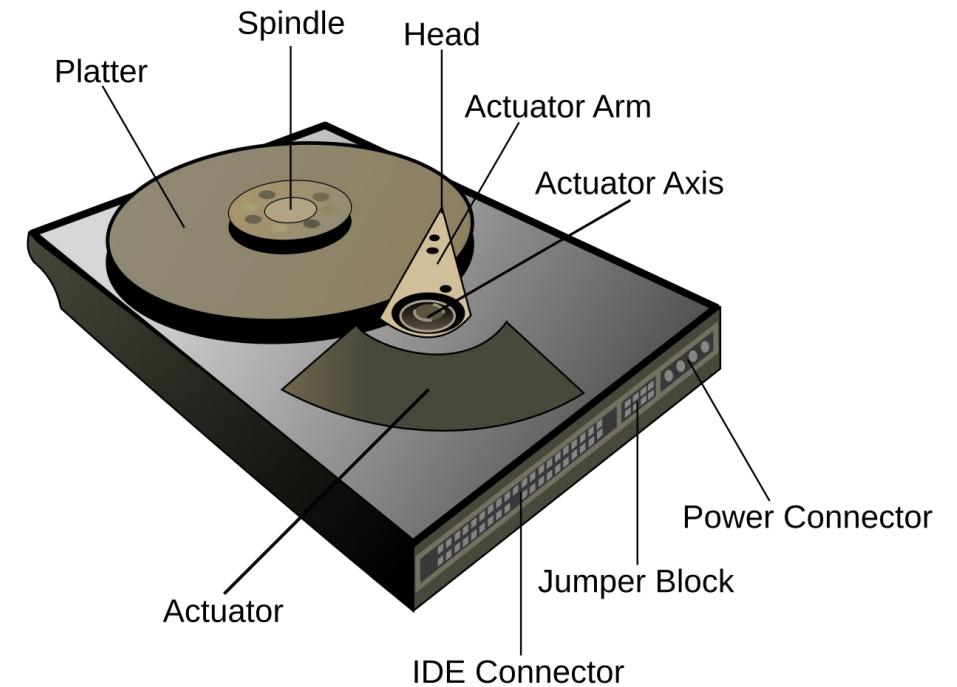
Storage devices

- Storage devices for non-volatile, long-time storage of data There was HDD disks replaced progressively by SSD disks



HDD disks

- **HDD (Hard Disk Drive):** a magnetic device, Older, uses spinning platters. Slower, cheaper for large capacities.
- use the same core technology as magneto phones
- Replaced by SSD disks.



SSD disk

- **SSD (Solid State Drive):** Newer, uses flash memory (like a USB drive). SSD drives are faster than HDD disks and more expensive.

